

(such as oak). Very hard woods will handle high loads comfortably and, more importantly, will not split at bolt attachment points. Soft woods like balsa should always be avoided. Also, while fastening the pieces of wood to other pieces, nuts and bolts should be used instead of screws to minimize the chance that the wood will be damaged or the holes will open up due to vibration.

Along with solid pieces of wood, you may also consider using a composite wood product like plywood, which consists of many sheets of wood glued together under pressure in such a way that its strength is augmented. Plywood is often manufactured from softer woods like spruce and the final product is much stronger than the sum of its constituents.

However you should not use composite materials like chipboard, as they cannot sustain much vibration and thus are inappropriate for robot body building purposes.

3.3.3 Plastics

Probably the number of plastic varieties available in the market is far greater than that of all varieties of other materials combined. Determining the correct type of plastic for a specific application can be a very arduous task and may take an unreasonably long period of time. Instead, if you want to work with plastics, it's better to pick a random variety which is available to you easily and start working on it, because all varieties of plastics are generally strong and tough, and thus are well suited for robot building applications.

Still, there are two most popular types of plastics you can consider:

Thermoset plastics

They are hard, tough and have a compactly meshed molecular structure. They are generally hardened in a mould using chemicals, and during this process they tend to release a lot of heat. This type of plastics can only be shaped by machining as heating them up will destroy the molecular structure and will make the material useless. Thermosets are generally used in hard plastic toys, appliances, furniture and other products which require high structural strength.

Elastomer plastics

They have a similar molecular structure to thermosets, but they have elastic characteristics also. Elastomers too, like thermosets, cannot be shaped up by heating.